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On a temporal model for the Chikungunya disease: Modeling, theory and numerics ☆

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Abstract

Reunion Island faced two episodes of Chikungunya, a vector-borne disease, in 2005 and in 2006. The latter was of unprecedented magnitude: one third of the population was infected. Until the severe episode of 2006, our knowledge of Chikungunya was very limited. The principal aim of our study is to propose a model, including human and mosquito compartments, that is associated to the time course of the first epidemic of Chikungunya. By computing the basic reproduction number R_0, we show there exists a disease-free equilibrium that is locally asymptotically stable if the basic reproduction number is less than 1. Moreover, we give a necessary condition for global asymptotic stability of the disease-free equilibrium. Then, we propose a numerical scheme that is qualitatively stable and present several simulations as well as numerical estimates of the basic reproduction number for some cities of Reunion Island. For the episode of 2005, R_0 was less than one, which partly explains why no outbreak appeared. Using recent entomological results, we investigate links between the episode of 2005 and the outbreak of 2006. Finally, our work shows that R_0 varied from place to place on the island, indicating that quick and focused interventions, like the destruction of breeding sites, may be effective for controlling the disease.
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1. Introduction

Chikungunya is transmitted by a domestic mosquito, like *Aedes albopictus* [16] and/or *Aedes aegypti* [9], which is also known to transmit Dengue. Chikungunya is endemic to many countries of Africa and Asia. It was first isolated in 1953 in Tanzania [30], but, following [6], it also existed before in Cairo and Batavia-Jakarta (1779), Zanzibar (1823 and 1870), India (1823, 1824–1825 and 1871–1872), Hong Kong, and Madras (1901–1902). Since the fifties, various outbreaks of Chikungunya have been reported in the Philippines (1954, 1956, and 1968) [5,21], Bangkok (1960s) [8], India (1964) [7,28], Senegal (1966, 1982, 1996) [11], Sri Lanka (1969), Vietnam (1975) and Indonesia (1973 and 1982) [22]. Then after a break of 20 years, outbreaks were observed in various Indian Ocean islands like Comoros, Mauritius, Mayotte and Reunion. And, since October 2006, the epidemic has appeared in several parts of India.

The virus emerged in Reunion Island in 2005: the first case was reported in January 2005. In 2005, no more than 5000 people were infected. But from the end of December 2005 till May 2006, the epidemic reached an unprecedented magnitude: one-third of the population was infected. It is now recognized that the principal vector of transmission of Chikungunya in Reunion Island is *Aedes albopictus*, also known as the *Asian tiger* mosquito. This mosquito is found in Southeast Asia, the Pacific and Indian Ocean islands, and up north through China and Japan. It recently appeared in Europe [23], and in the USA and Australia [4].

The symptoms of Chikungunya appear between 2 and 4 days after a bite by an infected mosquito: high fever and

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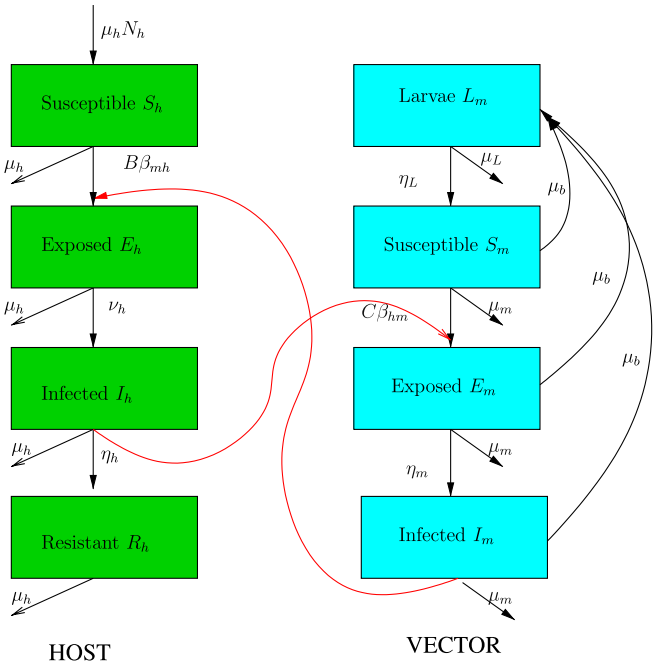


Fig. 1. A compartmental model for the Chikungunya disease.

headache occur, with arthritis affecting multiple joints (like ankles and wrist). Symptoms can persist several weeks or months (see [29] for further informations). A vaccine exists but is not commercially available yet, or in the near future. Infected people can be treated with drugs, but the efficacy of the treatment varies greatly from one person to another [29]. Thus, the main preventive measures are individual protection against mosquito bites and the control of the proliferation of mosquitoes, by reducing the number of breeding sites for instance.

Mathematics have become a major tool for studying the evolution of vector-borne diseases (see [1,12,18] and references therein). For instance, several temporal deterministic models have been proposed for Dengue (see [10,13,14]), but, to our knowledge, only very few for Chikungunya [3]. Reunion Island is now a very interesting place to study and compare models with respect to real data. In particular, the population of Reunion Island was a 'naive' population in 2005, which is one of the assumptions usually considered in mathematical models for epidemiologic studies. Reunion Island is in the south-west of the Indian Ocean and is famous for its chaotic landscape and more than a hundred microclimates. Climatic conditions can change rapidly from place to place. This is an important fact because mosquito's life-span is strongly related with the temperature and humidity.

In the present paper, we propose a temporal mathematical model to study the outbreak of 2005 in several cities of Reunion Island. Moreover, for epidemiological purposes, we give some estimates of the basic reproduction number R_0 [1,12,17,18]. For vector-borne disease, R_0 is defined as the number of secondary cases generated by a primary infectious case via the vectors in an entirely susceptible population. Thus, typically, if $R_0 < 1$ no outbreak appears while if $R_0 > 1$ an epidemic can occur. R_0 is classically used to quantify the intensity of interventions necessary to control an outbreak. We also provide data related to the disease free equilibrium, called here after *DFE*, and the endemic equilibrium, called here after *EE*. A non-standard numerical scheme is proposed that takes into account the properties of our differential system without restriction on the time-step. We present approximation of R_0 for some cities of Reunion Island and numerical simulations of the disease course, with comparison with real data. We finally give some conclusions and prospects.

2. A compartment model for the Chikungunya epidemic of 2005 – the reproduction number R_0

Transmission from human to human via the mosquito is the only way Chikungunya can spread. We assume homogeneous mixing between host and vector population, i.e. we assume that each vector has an equal probability to bite any host.

Our model (see Fig. 1) classifies hosts (the humans) into four epidemiological states: susceptible (S_h), exposed (E_h), infectious (I_h) and resistant (or immune, R_h). We assume that the total human population (N_h) is constant, i.e. $N_h = S_h + E_h + I_h + R_h$. The female mosquitoes are classified in four classes too: susceptible (S_m), exposed (E_m), infectious (I_m) and larval (L_m). The larval class includes the egg, larva and pupa stages. Both humans and mosquitoes are assumed to be born susceptible. The exposed (or incubating) classes, E_h and E_m, reflect the viral intrinsic and extrinsic incubation periods, $\frac{1}{\nu_h}$ days and $\frac{1}{\eta_m}$ days, respectively. The extrinsic incubation period is the time necessary for the virus to follow a cycle that brings it from the mosquito's stomach to its salivary gland. This incubation period can vary greatly depending, for example, on the temperature. For humans, the intrinsic incubation period or latent period is the period from the point of infection to the beginning of infectiousness. An infected human is infectious during $\frac{1}{\eta_h}$ days, called the viremic period, and then becomes resistant or immune.

Cross-infection between humans and vectors is modeled by the mass-action principle normalized by the total population of humans.

Every day, each mosquito bites, on average, B times. β_{mh} is the probability that a bite will lead to host infection. Thus $B\beta_{mh}$ represents the contact rate between infectious mosquitoes and susceptible hosts. We define $C\beta_{hm}$ in the same way: it is the contact rate between infectious hosts and susceptible mosquitoes. Furthermore, we suppose that infected mosquitoes have a fixed extrinsic incubation period that is followed by a definitive infectious state, i.e. infected vectors do not recover. The mean adult lifespan is $1/\mu_m$ days: we assume it is the same for all mosquitoes, i.e. the lifespan does not depend on whether the mosquito is infected or not.

Table 1
Epidemiological and entomological parameters

Parameters	Description	Mean value or range of values
B and C	Average daily biting (per day)	$\frac{1}{2}$ and 1
β_{mh}	Transmission probability from I_m (per bite)	[0.5;0.8]
β_{hm}	Transmission probability from I_h (per bite)	≈ 0.37 see [15]
$1/\mu_h$	Average lifespan of Humans (in days)	78×365
$1/\eta_h$	Mean viremic period (in days)	3 or 4
$1/\mu_m$	Average lifespan of adult mosquitoes (in days)	7
μ_b	Number of eggs at each deposit per capita (per day)	6
$1/\mu_L$	Natural mortality of larvae (per day)	[3; 4]
η_L	Maturation rate from larvae to adult (per day)	≈ 0.1
$1/\nu_m$	Extrinsic incubation period (in days)	3
$1/\nu_h$	Intrinsic incubation period (in days)	3 or 4

$$\begin{cases} \frac{dS_h}{dt}(t) = \mu_h N_h - B\beta_{mh}\frac{I_m}{N_h}S_h - \mu_h S_h \\ \frac{dE_h}{dt}(t) = B\beta_{mh}\frac{I_m}{N_h}S_h - \nu_h E_h - \mu_h E_h \\ \frac{dI_h}{dt}(t) = \nu_h E_h - \eta_h I_h - \mu_h I_h \\ \frac{dR_h}{dt}(t) = \eta_h I_h - \mu_h R_h \end{cases} \quad (1)$$

and

$$\begin{cases} \frac{dL_m}{dt}(t) = \mu_b\left(1-\frac{L_m}{K}\right)(S_m+E_m+I_m) - \eta_L L_m - \mu_L L_m \\ \frac{dS_m}{dt}(t) = -C\beta_{hm}\frac{I_h}{N_h}S_m - \mu_m S_m + \eta_L L_m \\ \frac{dE_m}{dt}(t) = C\beta_{hm}\frac{I_h}{N_h}S_m - \mu_m E_m - \eta_m E_m \\ \frac{dI_m}{dt}(t) = \eta_m E_m - \mu_m I_m \end{cases} . \quad (2)$$

Using the fact that $S_h + E_h + I_h + R_h$ is constant, equal to N_h, system (1)–(2) can be rewritten in the following way

$$\frac{dX}{dt} = A(X)X + F \quad (3)$$

with

$$X = (S_h, E_h, I_h, L_m, S_m, E_m, I_m)^T, A(X) = \begin{pmatrix} -B\beta_{mh}\frac{I_m}{N_h} - \mu_h & 0 & 0 & 0 & 0 & 0 & 0 \\ B\beta_{mh}\frac{I_m}{N_h} & -\nu_h - \mu_h & 0 & 0 & 0 & 0 & 0 \\ 0 & \nu_h & -\eta_h - \mu_h & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\left(\mu_L + \eta_L + \mu_b\frac{S_m+E_m+I_m}{K}\right) & \mu_b & \mu_b & \mu_b \\ 0 & 0 & 0 & \eta_L & -C\beta_{hm}\frac{I_h}{N_h} - \mu_m & 0 & 0 \\ 0 & 0 & 0 & 0 & C\beta_{hm}\frac{I_h}{N_h} & -\mu_m - \eta_m & 0 \\ 0 & 0 & 0 & 0 & 0 & \eta_m & -\mu_m \end{pmatrix}$$

The outbreak of 2005 appeared between March and June, that is at the end of the hot season and the beginning of the winter period. Thus, in our modeling, we assume that the mosquito population is maximal at the beginning of the outbreak and then decreases to a certain value. Moreover, we know that during the massive outbreak of 2006, the DRASS (a French government department of disease prevention and vector control) planned several interventions (like Kass' Moustik, see [20]) in order to reduce the number of breeding sites. We therefore decided to consider a supplementary compartment in the population of mosquitoes: the larval compartment, which includes the egg, larva and pupa stages. For the larval compartment, we assume there is no infection, i.e. we assume there is no vertical transmission of the virus. The oviposition rate per female is proportional to their density but, knowing that breeding sites can support only a limited number of larvae, pupae and eggs, we consider a maximal larval capacity effect, i.e. the oviposition rate is thus given by $\mu_b(1-\frac{L}{K})$, where K is the maximal larval capacity. The mortality rate for the larval stage is μ_L and η_L is the rate of transition (or maturation) from larva to adult. See also Table 1.

Altogether, we obtain the following systems of equations and $F = (\mu_h N_h, 0, 0, 0, 0, 0, 0)^T$. Note that $A(X)$ is a Metzler matrix, i.e. a matrix such that off diagonal terms are non-negative, for all $X \in \mathbb{R}^7_+$. Thus, using the fact that $F \geqslant 0$, system (3) is positively invariant in $\mathbb{R}^7_+$, which means that any trajectory of the system starting from an initial state in the positive orthant $\mathbb{R}^7_+$ remains forever in $\mathbb{R}^7_+$. The right-hand side is Lipschitz continuous: there exists an unique maximal solution.

Let $\mathcal{G} = \{(S_h, E_h, I_h, L_m, S_m, E_m, I_m) \in \mathbb{R}^7_+ / S_h \leqslant N_h, E_h \leqslant N_h, I_h \leqslant N_h, L_m \leqslant K, S_m \leqslant \frac{\eta_L}{\mu_m}K, E_m \leqslant \frac{\eta_L}{\mu_m}K, I_m \leqslant \frac{\eta_L}{\mu_m}K, S_h + E_h + I_h \leqslant N_h$ and $S_m + E_m + I_m \leqslant \frac{\eta_L}{\mu_m}K\}$. It can be verified that $\mathcal{G}$ is positively invariant with respect to (3).

In Appendix A, we show that if $\mu_b > \frac{\mu_m(\eta_L+\mu_L)}{\eta_L}$, the disease-free equilibrium of (1)–(2), called here-after *DFE*, is $(N_h, 0, 0, 0, L_{m0}, S_{m0}, 0, 0)$. If $\mu_b \leqslant \frac{\mu_m(\eta_L+\mu_L)}{\eta_L}$, the disease-free equilibrium is $(N_h, 0, 0, 0, 0, 0, 0, 0)$, which is not biologically realistic because in a tropical island like Reunion Island it seems not realistic to suppose that mosquito populations could completely vanish (anyway, in Appendix A, we give a very short study related to this particular *DFE*).

Moreover, if $\mu_b > \frac{\mu_m(\eta_L+\mu_L)}{\eta_L}$, there exists an endemic equilibrium of (1)–(2), i.e. with $I_h > 0$ and $I_m > 0$, $EE = (S^*_h, E^*_h, I^*_h, R^*_h, L_{m0}, S^*_m, E^*_m, I^*_m)$ (see Appendix A for further details). Thus, in the rest of the paper, we will assume that $\mu_b > \frac{\mu_m(\eta_L+\mu_L)}{\eta_L}$.

2.1. The reproduction number R_0

An easy way to theoretically compute the reproduction number is to follow [31]. In system (1)–(2), we consider only the terms in which the infection is in progression, i.e. E_h, I_h, E_m and I_m. The corresponding equations can be rewritten in the following way

$$\dot{x}_i = f_i(x) = \mathcal{F}_i(x) - \left(\mathcal{V}_i^-(x) - \mathcal{V}_i^+(x)\right), \quad i = 1, \ldots, 4 \tag{4}$$

where $\mathcal{F}_i$ represents the rate of appearance of new infections in compartment i, $\mathcal{V}_i^+$ is the rate of transfer of individuals into compartment i by all other means, and $\mathcal{V}_i^-$ is the rate of transfer of individuals out of compartment i. Thus we obtain

$$\begin{cases} \frac{dE_h}{dt}(t) = B\beta_{mh}\frac{I_m}{N_h}S_h - \nu_h E_h - \mu_h E_h \\ \frac{dI_h}{dt}(t) = \nu_h E_h - \eta_h I_h - \mu_h I_h \\ \frac{dE_m}{dt}(t) = C\beta_{hm}\frac{I_h}{N_h}S_m - \mu_m E_m - \eta_m E_m \\ \frac{dI_m}{dt}(t) = \eta_m E_m - \mu_m I_m \end{cases} \tag{5}$$

that can be rewritten as $\frac{dx}{dt} = \mathcal{F}(x) - \mathcal{V}(x)$, where $x^T = (E_h, I_h, E_m, I_m)$ and

$$\mathcal{F}(x_I) = \begin{pmatrix} B\beta_{mh}\frac{I_m}{N_h}S_h \\ 0 \\ C\beta_{hm}\frac{I_h}{N_h}S_m \\ 0 \end{pmatrix},$$

$$\mathcal{V}(x) = \begin{pmatrix} (\nu_h + \mu_h)E_h \\ -\nu_h E_h + (\eta_h + \mu_h)I_h \\ (\mu_m + \eta_m)E_m \\ -\eta_m E_m + \mu_m I_m \end{pmatrix}.$$

In [31], the authors showed that the general basic reproduction number is given by $\mathcal{R}_0 = \rho(J_{\mathcal{F}}J_{\mathcal{V}}^{-1})$, where $\rho(A)$ denotes the spectral radius of A. $J_{\mathcal{F}}$ and $J_{\mathcal{V}}$ are the Jacobian matrices associated with $\mathcal{F}$ and $\mathcal{V}$ and describe the linearization of the reduced system around *DFE*. We obtain

$$R_0^2 = \frac{\eta_m \nu_h B C \beta_{hm}\beta_{mh}}{(\nu_h + \mu_h)(\eta_h + \mu_h)\mu_m(\mu_m + \eta_m)}\frac{S_{m0}}{N_h}. \tag{6}$$

See Appendix B for more details.

Furthermore, following [31], we know that if $R_0 < 1$, *DFE* is locally stable, and it is locally unstable if $R_0 > 1$, which means that the introduction of a virus can lead to an outbreak. Finally, we summarize as follows:

Theorem 1. *If* $\mu_b > \frac{\mu_m(\eta_L+\mu_L)}{\eta_L}$, $(N_h, 0, 0, 0, L_{m0}, S_{m0}, 0, 0)$ *is the disease-free equilibrium of system (1)–(2), which is locally asymptotically stable if* $R_0 < 1$ *and locally asymptotically unstable if* $R_0 > 1$.

Remark 2. Let us now explain the biological meaning of R_0 from the dimensional parameters. The term $R_{mh} = \frac{B\beta_{mh}}{\mu_m}\frac{\nu_h}{\mu_h+\nu_h}$, in the square root represents the disease R_0 from mosquitoes to humans as the product between the transmission probability $B\beta_{mh}$, the adult mosquito (infectious) life span $\frac{1}{\mu_m}$ and the proportion $\frac{\nu_h}{\mu_h+\nu_h}$ of humans that survive the incubation period. The other term in the square root, $R_{hm} = \frac{1}{\eta_h+\mu_h}\frac{\eta_m}{(\mu_m+\eta_m)}C\beta_{hm}\frac{S_{m0}}{N_h}$, gives R_0 from humans to mosquitoes, that is the product between the transmission probability $C\beta_{hm}$ multiplied by the number of susceptible mosquitoes per human at equilibrium $\frac{S_{m0}}{N_h}$, the human's viremic period $\frac{1}{\eta_h+\mu_h}$ and the proportion $\frac{\eta_m}{\mu_m+\eta_m}$ of mosquitoes that survive the incubation period. Finally, the product gives the total disease from vector to vector or human to human. The square root comes from the 'two generations' required for an infected vector or host to reproduce itself. Thus, we have

$$R_0^2 = R_{mh} \times R_{hm}. \tag{7}$$

Note also that the parameters of the larval stage influence R_{hm} indirectly and thus R_0, through S_{m0}.

Remark 3. There exists a unique endemic equilibrium *EE* since $R_0 > 1$. For the moment, we have not planned an extensive study of *EE* because the system is difficult and it is not the aim of this study.

2.2. Global asymptotic stability of DFE

We used a result obtained by Kamgang and Sallet [19], which is an extension of some results given in [31]. Using the property of *DFE*, it is possible to rewrite (1)–(2) in the following manner

$$\begin{cases} \dot{x}_S = A_1(x)(x_S - x_{DFE,S}) + A_{12}(x)x_I \\ \dot{x}_I = A_2(x)x_I \end{cases}, \tag{8}$$

where x_S is the vector representing the state of different compartments of non-transmitting individuals (e.g. susceptible, immune) and the vector x_I represents the state of compartments of different transmitting individuals (e.g. infected, exposed). Here, we have $x_S = (S_h, R_h, L_m, S_m)^T$, $x_I = (E_h, I_h, E_m, I_m)^T$, $x_{DFE,S} = (N_h, 0, L_{m0}, S_{m0})^T$. Indeed, from

$$\begin{aligned} \frac{dS_h}{dt}(t) &= -\mu_h(S_h - N_h) - B\beta_{mh}\frac{I_m}{N_h}S_h \\ &= -\mu_h(S_h - S_{h0}) - B\beta_{mh}\frac{I_m}{N_h}S_h \\ \frac{dR_h}{dt}(t) &= \eta_h I_h - \mu_h R_h, \end{aligned}$$

we obtain the first and second lines of $A_1(x)$ and $A_{12}(x)$. Then, from

$$\begin{aligned} \frac{dL_m}{dt}(t) &= \mu_b\left(1 - \frac{L_m}{K}\right)(S_m + E_m + I_m) - \eta_L L_m - \mu_L L_m \\ \frac{dS_m}{dt}(t) &= -C\beta_{hm}\frac{I_h}{N_h}S_m - \mu_m S_m + \eta_L L_m, \end{aligned}$$

and using the relation $(\eta_L+\mu_L)L_{m0}=\mu_b(1-\frac{L_{m0}}{K})S_{m0}$ (see Appendix A), we obtain

$$\begin{aligned}\frac{dL_m}{dt}(t) &= \mu_b\left(1-\frac{L_m}{K}\right)(S_m+E_m+I_m)-(\eta_L+\mu_L)\\ &\quad\times(L_m-L_{m0})-\mu_b\left(1-\frac{L_{m0}}{K}\right)S_{m0}\\ &= \mu_b\left(1-\frac{L_m}{K}\right)(E_m+I_m)-\mu_b\frac{L_m}{K}S_m-(\eta_L+\mu_L)\\ &\quad\times(L_m-L_{m0})+\mu_b(S_m-S_{m0})+\mu_b\frac{L_{m0}}{K}S_{m0}\\ &= -\left(\eta_L+\mu_L+\mu_b\frac{S_m}{K}\right)(L_m-L_{m0})+\mu_b\left(1-\frac{L_{m0}}{K}\right)\\ &\quad\times(S_m-S_{m0})+\mu_b\left(1-\frac{L_m}{K}\right)(E_m+I_m).\end{aligned}$$

Thus, we obtain the following matrices for $A_1(x)$, $A_{12}(x)$ and $A_2(x)$

$$A_1(x)=\begin{pmatrix}-\mu_h & 0 & 0 & 0\\ 0 & -\mu_h & 0 & 0\\ 0 & 0 & -\left(\mu_L+\eta_L+\mu_b\frac{S_m}{K}\right) & \mu_b\left(1-\frac{L_{m0}}{K}\right)\\ 0 & 0 & \eta_L & -\mu_m\end{pmatrix},$$

$$A_{12}(x)=\begin{pmatrix}0 & 0 & 0 & -B\beta_{mh}\frac{S_h}{N_h}\\ 0 & \eta_h & 0 & 0\\ 0 & 0 & \mu_b\left(1-\frac{L_m}{K}\right) & \mu_b\left(1-\frac{L_m}{K}\right)\\ 0 & -C\beta_{hm}\frac{S_m}{N_h} & 0 & 0\end{pmatrix}$$

and

$$A_2(x)=\begin{pmatrix}-(\nu_h+\mu_h) & 0 & 0 & B\beta_{mh}\frac{S_h}{N_h}\\ \nu_h & -(\eta_h+\mu_h) & 0 & 0\\ 0 & C\beta_{hm}\frac{S_m}{N_h} & -(\mu_m+\eta_m) & 0\\ 0 & 0 & \eta_m & -\mu_m\end{pmatrix}.$$

A direct computation shows that the eigenvalues of $A_1(x)$ are real and negative. Thus the system $\dot{x}_S=A_1(x)(x_S-x_{DFE,S})$ is globally asymptotically stable (GAS) at $x_{DFE,S}$. Note also that $A_2(x)$ is a Metzler matrix. In fact $A_2(x)$ is the matrix we used in the previous section to compute the next generation matrix and the basic reproduction number R_0^2.

Following $\mathcal{G}$, we now consider the bounded set $\mathcal{B}$:

$$\begin{aligned}\mathcal{B}=\Big\{&(S_h,R_h,L_m,S_m,E_h,I_h,E_m,I_m)\in\mathbb{R}_+^8/S_h\leqslant N_h,\\ &E_h\leqslant N_h,I_h\leqslant N_h,R_h\leqslant N_h,L_m\leqslant K,S_m\leqslant\frac{\eta_L}{\mu_m}K,\\ &E_m\leqslant\frac{\eta_L}{\mu_m}K,I_m\leqslant\frac{\eta_L}{\mu_m}K,S_h+E_h+I_h+R_h=N_h\\ &\text{and}\quad S_m+E_m+I_m\leqslant\frac{\eta_L}{\mu_m}K\Big\}.\end{aligned}$$

Let us recall the following theorem [19]:

Theorem 4. *Let $\mathcal{B}\subset\mathcal{U}=\mathbb{R}_+^4\times\mathbb{R}_+^4$. The system (8) is of class C^1, defined on $\mathcal{U}$. If*

(1) *$\mathcal{B}$ is positively invariant relative to (8).*
(2) *The system $\dot{x}_S=A_1(x)(x_S-x_{DFE,S})$ is GAS at $x_{DFE,S}$.*
(3) *For any $x\in\mathcal{B}$, the matrix $A_2(x)$ is Metzler irreducible.*
(4) *There exists a matrix $\overline{A}_2$, which is an upper bound of the set $\mathcal{M}=\{A_2(x)\in\mathcal{M}_4(\mathbb{R})|x\in\overline{\mathcal{B}}\}$, with the property that if $\overline{A}_2\in\mathcal{M}$, for any $\bar{x}\in\overline{\mathcal{B}}$, such that $A_2(\bar{x})=\overline{A}_2$, then $\bar{x}\in\mathbb{R}^4\times\{0\}$.*
(5) *The stability modulus of $\overline{A}_2$, $\alpha(\overline{A}_2)=\max_{\lambda\in Sp(A_2)}\mathcal{R}e(\lambda)$, satisfies $\alpha(\overline{A}_2)\leqslant 0$.*

Then DFE is GAS in $\overline{G}$.

Proof. See [19] for a proof in a more general setting. Let us now verify the assumptions of the previous theorem: it is obvious that conditions (1–3) of the theorem are satisfied. In particular, for all $x\in\mathcal{B}$, $A_2(x)$ is irreducible because $(1+|A_2(x)|)^3>0$. An upper bound of the set of matrices $\mathcal{M}$, which is the matrix $\overline{A}_2$, is given by $A_2(\bar{x})$, where $\bar{x}=(N_h,0,\overline{S}_m,\overline{L}_m,0,0,0,0)\in\mathbb{R}^4\times\{0\}$, where $\overline{S}_m=\frac{\eta_L}{\mu_m}K$ and $\overline{L}_m=K$. Matrix $\overline{A}_2$ is irreducible for the same reason as for matrices $A_2(x)$. □

To check condition (5) in the theorem, we will use the following useful lemma [19]:

Lemma 5. *Let M be a square Metzler matrix written in block form $M=\begin{pmatrix}A & B\\ C & D\end{pmatrix}$, with A and D square matrices. M is Metzler stable if and only if matrices A and $D-CA^{-1}B$ are Metzler stable.*

In our case, we have $A=\begin{pmatrix}-(\nu_h+\mu_h) & 0\\ \nu_h & -(\eta_h+\mu_h)\end{pmatrix}$, $B=\begin{pmatrix}0 & B\beta_{mh}\\ 0 & 0\end{pmatrix}$, $C=\begin{pmatrix}0 & C\beta_{hm}\frac{\overline{S}_m}{N_h}\\ 0 & 0\end{pmatrix}$, and $D=\begin{pmatrix}-(\mu_m+\nu_m) & 0\\ \eta_m & -\mu_m\end{pmatrix}$. Clearly, A is a stable Metzler matrix. Then, after some computations, we obtain

$$D-CA^{-1}B=\begin{pmatrix}-(\mu_m+\nu_m) & \frac{\nu_hB\beta_{mh}C\beta_{hm}}{(\eta_h+\mu_h)(\nu_h+\mu_h)}\frac{\overline{S}_m}{N_h}\\ \eta_m & -\mu_m\end{pmatrix}.$$

$D-CA^{-1}B$ is a stable Metzler matrix if $\mu_m(\mu_m+\eta_m)-\frac{\eta_m\nu_hB\beta_{mh}C\beta_{hm}}{(\eta_h+\mu_h)(\nu_h+\mu_h)}\frac{\overline{S}_m}{N_h}\geqslant 0$, that is if $1-R_C^2\geqslant 0$ with

$$R_C^2=\frac{\eta_m\nu_hBC\beta_{hm}\beta_{mh}}{(\nu_h+\mu_h)(\eta_h+\mu_h)\mu_m(\mu_m+\eta_m)}\frac{\eta_L}{\mu_m}\frac{K}{N_h}. \tag{9}$$

This is a necessary condition for *DFE* to be GAS. Note that $R_0^2=\frac{\mu_m\overline{S}_{m0}}{\eta_LK}R_C^2$. Since the value of the matrix $\overline{A}_2$ is not attained to *DFE*, the necessary condition for GAS we obtain is not R_0. But if we compute $\alpha(A_2(DFE))$ by applying the previous lemma we recover R_0.

Finally, from Theorem 4 and Lemma 5, we deduce the following:

Theorem 6. *If* $\mu_b > \frac{\mu_m(\eta_L+\mu_L)}{\eta_L}$, $(N_h, 0, 0, 0, L_{m0}, S_{m0}, 0, 0)$ *is the disease-free equilibrium of system (1)–(2), which is globally asymptotically stable if* $R_C \leqslant 1$.

Remark 7. This latter result is of outmost importance, because it shows that if at any time, through appropriate interventions (e.g. destruction of breeding sites, use of insecticides and repellants), we are able to lower R_0 and R_C to less than 1, then the disease will disappear. In fact, since $R_C > 1$ and $R_0 < 1$ in 2005 (see Table 3), no outbreak appeared: $R_C > 1$ indicates that an epidemic could occur, while $R_0 < 1$ shows that since the mosquitoes population decreases to the equilibrium, the outbreak will stop.

We know that R_0 and R_C depend on K, the breeding capacity. Thus, if we are able to lower K, R_0 and R_C will decrease. This shows that it was important to take into account the larvae stage in the model.

Remark 8. After the huge episode of 2006, the DRASS carried out several interventions, like the destruction of breeding sites, and these can partly explain why no more outbreaks appeared. Only a few cases were reported from time to time.

3. The numerical scheme

In the previous section, we presented different properties of the theoretical solution that we may take into account in our numerical scheme. Compartmental models are usually solved using standard numerical integrators, e.g. Euler or Runge Kutta methods. These methods, however, can lead to spurious solutions or exhibit numerical instabilities for certain values of the time discretization or the model parameters. Here, we aim to construct a robust scheme that preserves the positivity of our solutions as well as the relation $S_h + E_h + I_h + R_h = N_h$. Moreover, we require our scheme to be qualitatively stable in the sense of [2], that is the numerical solution should reproduce any property of the exact solution. To do that, we follow Mickens' rules to obtain a non-standard numerical scheme (see [24–27] and references therein). In particular, we use Mickens' second rule: we approximate the non-linear terms in a non-local way and consider some linear terms in an implicit way. Thus the approximation scheme for the system (1)–(2) is given by

$$\begin{cases} \frac{S_h^{n+1}-S_h^n}{\Delta t} = \mu_h N_h - B\beta_{mh}\frac{I_m^n}{N_h}S_h^{n+1} - \mu_h S_h^{n+1} \\ \frac{E_h^{n+1}-E_h^n}{\Delta t} = B\beta_{mh} I_m^n \frac{I_m^n}{N_h} S_h^{n+1} - (v_h+\mu_h)E_h^{n+1} \\ \frac{I_h^{n+1}-I_h^n}{\Delta t} = v_h E_h^{n+1} - (\eta_h+\mu_h)I_h^{n+1} \\ \frac{R_h^{n+1}-R_h^n}{\Delta t} = \eta_h I_h^{n+1} - \mu_h R_h^{n+1} \\ \frac{L_m^{n+1}-L_m^n}{\Delta t} = \mu_b(S_m^n + I_m^n + R_m^n) \\ \quad -(\mu_L + \eta_L + \frac{\mu_b}{K}(S_m^n + I_m^n + R_m^n))L_m^{n+1} \\ \frac{S_m^{n+1}-S_m^n}{\Delta t} = -C\beta_{hm}\frac{I_h^n}{N_h}S_m^{n+1} - \mu_m S_m^{n+1} + \eta_L L_m^{n+1} \\ \frac{E_m^{n+1}-E_m^n}{\Delta t} = C\beta_{hm}\frac{I_h^n}{N_h}S_m^{n+1} - (v_m+\mu_m)E_m^{n+1} \\ \frac{I_m^{n+1}-I_m^n}{\Delta t} = v_m E_m^{n+1} - \mu_m I_m^{n+1} \end{cases},$$

where Δt is the time-step, S_h^n, E_h^n, I_h^n, R_h^n, S_m^n, E_m^n and I_m^n are approximations of $S_h(t_n)$, $E_h(t_n)$, $I_h(t_n)$, $R_h(t_n)$, $S_m(t_n)$, $E_m(t_n)$ and $I_m(t_n)$. Rearranging the previous formulations, we obtain

$$\begin{cases} S_h^{n+1} = \frac{S_h^n + \Delta t\mu_h N_h}{1+\Delta t(B\beta_{mh}I_m^n/N_h + \mu_h)} \\ E_h^{n+1} = \frac{E_h^n + \Delta t B\beta_{mh} I_m^n S_h^{n+1}/N_h}{1+\Delta t(v_h+\mu_h)} \\ I_h^{n+1} = \frac{I_h^n + \Delta t v_h E_h^{n+1}}{1+\Delta t(\eta_h+\mu_h)} \\ R_h^{n+1} = \frac{R_h^n + \Delta t\eta_h I_h^{n+1}}{1+\Delta t\mu_h} \end{cases} \tag{10}$$

and

$$\begin{cases} L_m^{n+1} = \frac{L_m^n + \Delta t\mu_b(S_m^n + I_m^n + R_m^n)}{1+\Delta t(\mu_L+\eta_L+\frac{\mu_b}{K}(S_m^n+I_m^n+R_m^n))} \\ S_m^{n+1} = \frac{S_m^n + \Delta t\eta_L L_m^{n+1}}{1+\Delta t(C\beta_{hm}I_h^n/N_h+\mu_m)} \\ E_m^{n+1} = \frac{E_m^n + \Delta t C\beta_{hm} I_h^n S_m^{n+1}/N_h}{1+\Delta t(v_m+\mu_m)} \\ I_m^{n+1} = \frac{I_m^n + \Delta t v_m E_m^{n+1}}{1+\Delta t\mu_m} \end{cases}. \tag{11}$$

Since the right-hand sides of the previous equations are positive, it is clear that, for any positive initial data, we will obtain positive approximations that are located in the feasible region. Moreover it is easy to check that $S_h^{n+1} + E_h^{n+1} + I_h^{n+1} + R_h^{n+1} = S_h^n + E_h^n + I_h^n + R_h^n$ for all $n \in \mathbb{N}$, which implies that $S_h^n + E_h^n + I_h^n + R_h^n = N_h$, for all $n \in \mathbb{N}$.

It is easy to verify that *DFE* is a fixed point of the numerical algorithm. In Appendix C, we prove that the stability/instability property of *DFE* is preserved by our numerical scheme, for all time steps $\Delta t > 0$. In particular, we have

Theorem 9. *Since* $R_0 < 1$ *the numerical scheme will converge to the disease-free equilibrium DFE for every* $\Delta t > 0$.

4. Numerical simulations and comparison with real data

We performed numerical simulations for some cities for which we obtained epidemiological data: Saint-Denis, the capital of Reunion Island, located in the north, La Possession and Le Port, located in the north-west of the Island, and Saint-Pierre, located in the south-south-west. All these cities are at the sea level. Moreover we compared our simulations with real data, corresponding to declared cases recorded since the beginning of the outbreak by the DRASS in cooperation with the CIRE (French government health services) through their survey networks.

4.1. Parameters and simulations

Table 1 shows the parameters used in our computations. In this table, some parameters can change from place to place. Most of the values were obtained from entomologists and are related to experiments on *Aedes albopictus* conducted by Dr. H. Delatte (IRD, France) and Dr. A.

Failloux (Institut Pasteur, France). Note also that we always verified $\mu_b \geqslant \frac{(\eta_L+\nu_L)\mu_m}{\eta_L}$ which ensured the existence of a biologically realistic DFE.

In the following computations (see Figs. 2–5) we consider that at the beginning of each episode ($t = 0$ in our figures), there are m female mosquitoes per human, i.e. the whole population of female mosquitoes is thus $S_m(0) = m \times N_h$. In the same way, for the maximal capacity K, we consider a number k of larvae per human. Thus K is given by $K = k \times N_h$ and we choose $L_m(0) = k \times N_h$. In Table 2 we summarize the initial values used in the computations for each city.

Table 2 shows k values for the numerical simulation. To choose k, we use the Breteau Index, corresponding to the number of breeding sites per one hundred houses. In 2005 and 2006, the DRASS measured the Breteau's index in different cities of Reunion Island. For instance, because the index was much higher (around 60) in Saint-Denis than in the three others, we choose $k = 3$.

In Table 3 are summarized the results obtained for R_0^2, R_{mh} and R_{hm} in each city.

Because no epidemic appeared in 2005, it is not surprising that the values for R_0^2 are less than one. Of particular interest is R_{hm} (see relation (7)), which represents the disease transmission from humans to mosquitoes: our numerics show that R_{hm} can vary from place to place, but because it is small in comparison with R_{mh}, we get $R_0^2 < 1$. This result points out the great influence of climate parameters (e.g. temperature, humidity) on the mosquito's biology. Knowing that Reunion Island has more than one hundred micro-climates, it is clear that the time course of the disease can change from place to place. Thus, we need a local study that takes into account specific parameters of the area of interest. For instance, the mean daily temperature in the city of Le Port is higher than in other cities in Reunion Island, thus we consider $m = 9$ in Table 2.

The numerics only give a mean behavior of the time course of the disease; hence, it is impossible to fit the real

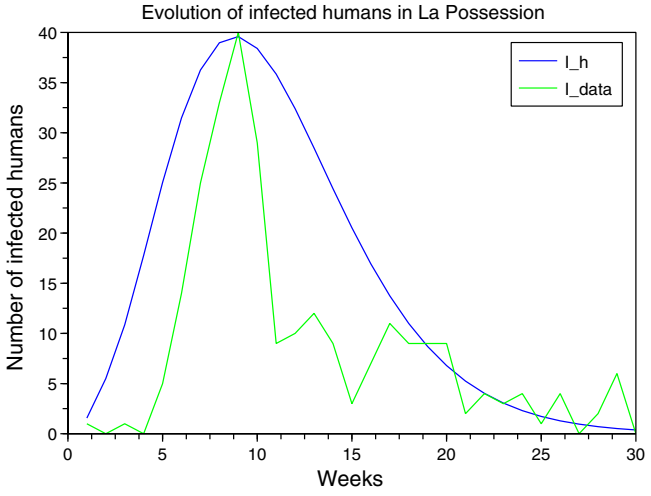


Fig. 3. Evolution of the infected population per week in La Possession; comparison with real data.

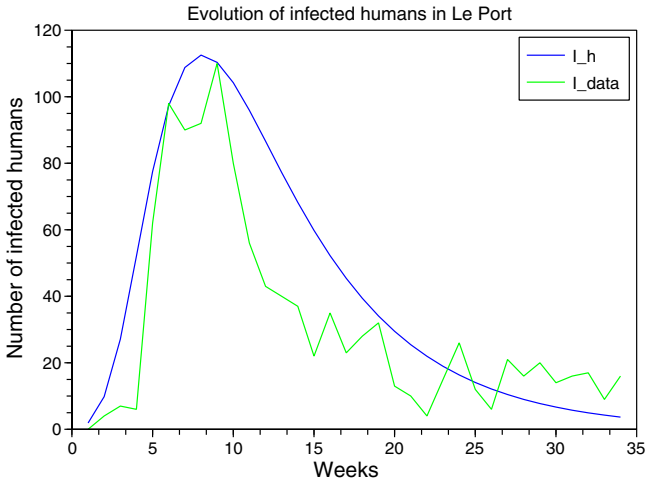


Fig. 4. Evolution of the infected population per week in Le Port; comparison with real data.

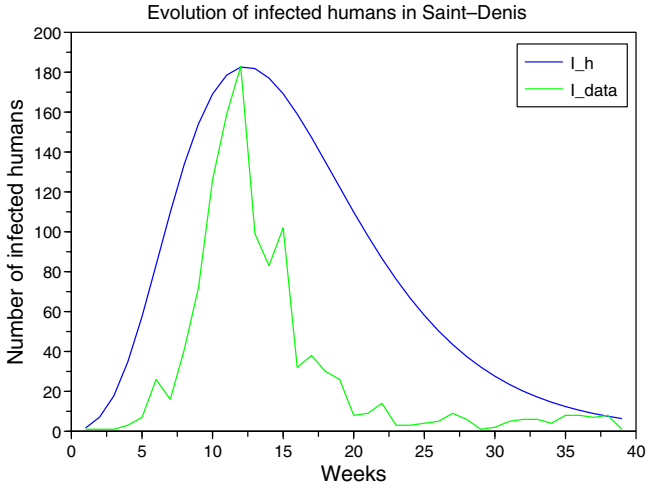


Fig. 2. Evolution of the infected population per week in Saint-Denis; comparison with real data.

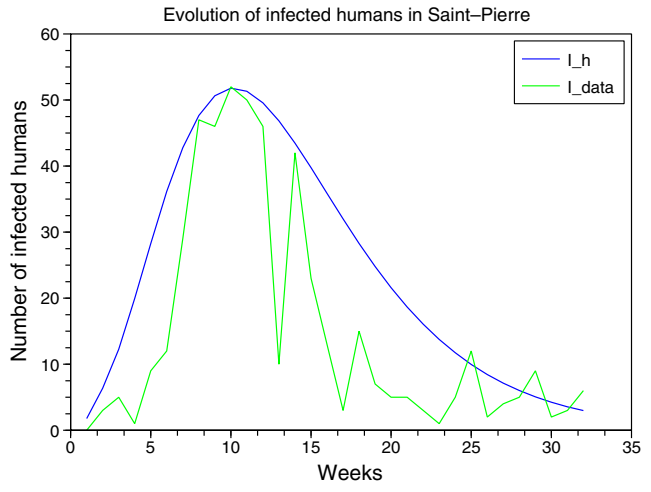


Fig. 5. Evolution of the infected population per week in Saint-Pierre; comparison with real data.

Table 2
Initial values for each city

City	N_h	m	k
Saint-Denis	136 000	6	3
La Possession	24 500	4	3
Le Port	40 000	9	1
Saint-Pierre	73 000	3	2

Table 3
R_0^2 and R_C^2 for some cities in 2005

Towns	R_0^2	R_{mh}	R_{hm}	R_C^2
Saint-Denis	0.84	2.614	0.324	6.42
La Possession	0.347	2.55	0.135	4.16
Le Port	0.91	2.834	0.322	2.53
Saint-Pierre	0.59	2.58	0.227	4.13

Table 4
Basic reproduction numbers for some cities in December 2005

Towns	R_0^2	R_{mh}	R_{hm}
Saint-Denis	2.12	2.614	0.811
La Possession	0.89	2.55	0.35
Le Port	2.28	2.834	0.806
Saint-Pierre	1.516	2.59	0.585

values. According to our model the disease should disappear after a while, but this is not what happened in reality.

Indeed, the disease survived the winter period and rose again in December 2005–January 2006, at the beginning of the hot period. This survival can be explained by many facts: a mild winter that allowed a lot of mosquitoes to survive, vertical transmission of the virus, etc. Anyway, it seems surprising that, with the values obtained for R_0 in Table 3 an epidemic occurred again from December 2005 till May 2006. Many explanations have been proposed. From November till March, there is the rain season in Reunion Island and then it is easy to imagine that climatic conditions enhance the number of breeding sites which leads to a larger population of susceptible mosquitoes. At the same time, the capacity of the breeding sites, K, increases and so does the basic reproduction number R_0. Another important factor is some change in the virus: recent experiments [15] show that β_{hm}, the rate of transmission from humans to mosquitoes, suddenly increased from 0.37 to 0.95 at the end of 2005. If we consider this new value in (6) we obtain new estimates for R_0 in December 2005 (see Table 4), that are greater than one, except in La Possession, which can lead to an epidemic in some places. We believe that the conjunction of many factors led to the epidemic of 2006.

This result about the variation of R_0 is important because it shows that in some places, interventions for vector control must be immediate and strong. At the beginning of an epidemic, it is important to localize places where the basic reproduction number has a good chance of being large. To generalize our results, we need further informations about the entomological parameters (currently carried out in the ANR project 'Entomochik') and, of course, epidemiological data for other cities in Reunion Island.

5. Conclusion

To our knowledge, this is the first time a larval compartment with maximal larval capacity is introduced in models of the evolution of the Chikungunya epidemic. From the theoretical point of view, we have showed that the disease free equilibrium is locally asymptotically stable since $R_0 < 1$, which is a very important result for epidemiologists (and even for entomologists) seeking to control a disease (via the control of the vector population). In particular, our results show that if at any time R_C is less than or equal to 1, then the disease will disappear. Following Mickens rules, we have proposed a non-standard numerical scheme that reproduces the properties of the differential system for any time-step $\Delta t > 0$. Then we have performed numerical simulations and have compared them to real data. The basic reproduction number changed during 2005: from less than 1 to greater than 1, which agrees with the observed situation. Moreover, the values computed for R_0 show that the force of infection can vary from place to place. This is very important for setting up interventions quickly, such as destruction of breeding sites in places where we know that the force of infection is known to be higher than in others.

One of the next steps would be to generalize our study to other cities in Reunion Island, especially during the episode of 2006, in order to get a map of the basic reproduction number. Another step for further studies would be to continue our investigations on the relation between the outbreak of 2005 and the epidemic of 2006. In particular, it would be interesting to take into account in our model the effect of larviciding or the destruction of breeding sites in order to show if these actions can be efficient to control or even to stop the epidemic. In the paper, we have begun to give a preliminary explanation. An attempt was made in [3], where the author considered a model for the whole Island, with periodicity in the mosquito population. Unfortunately, in this study, the episode of 2005 was not well fitted and the author did not study the long term epidemic evolution of the model and the risk for the population. Finally, it would be of interest to link the parameters used in our model with weather parameters, like temperature and humidity. Such relations exist for *Aedes aegypti* but not yet for *Aedes albopictus*.

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the University of Pretoria (South Africa), where a part of this work was completed, for their kind hospitality.

Appendix A

We compute the disease-free equilibrium *DFE* and the endemic equilibrium *EE* associated with system (1)–(2). We solve (1) with $\frac{dS_h}{dt}(t) = 0 = \frac{dE_h}{dt}(t) = \frac{dI_h}{dt}(t) = \frac{dR_h}{dt}(t) = 0$, and we obtain the following relations

$$\begin{cases} E_h = \frac{\mu_h+\eta_h}{\nu_h} I_h \\ R_h = \frac{\eta_h}{\mu_h} I_h \\ S_h = N_h - \frac{(\mu_h+\nu_h)(\mu_h+\eta_h)}{\mu_h \nu_h} I_h \\ \frac{I_h}{N_h} = \frac{\nu_h \mu_h B \beta_{mh} I_m}{(\mu_h+\nu_h)(\mu_h+\eta_h)(\mu_h N_h + B\beta_{mh} I_m)} \end{cases}. \tag{12}$$

Then, we solve (2) with $\frac{dL_m}{dt}(t) = 0 = \frac{dS_m}{dt}(t) = \frac{dE_m}{dt}(t) = \frac{dI_m}{dt}(t) = 0$ and we obtain the following relations

$$\begin{cases} E_m = \frac{\mu_m}{\eta_m} I_m \\ L_m = \frac{\mu_m}{\eta_L}(S_m + E_m + I_m) \\ (\eta_L + \mu_L) L_m = \mu_b (1 - \frac{L_m}{K}) \frac{\eta_L}{\mu_m} L_m \end{cases}.$$

From the last equality, we deduce that either $L_{m0} = 0$ or $L_{m0} = \left(1 - \frac{(\eta_L+\mu_L)\mu_m}{\mu_b \eta_L}\right) K$ if $\mu_b > \frac{(\eta_L+\mu_L)\mu_m}{\eta_L}$. If $L_m = 0$, then $S_m = E_m = I_m = 0$ and this implies that $I_h = E_h = R_h = 0$ and $S_h = N_h$. If $L_m > 0$, then the *DFE* is such that $I_m = I_h = 0$, which implies, from the previous relation, that $E_m = E_h = R_h = 0$, $S_{m0} = (\frac{\eta_L}{\mu_m}\mu_b - (\eta_L + \mu_L))K$, and $S_h = N_h$. Finally, if $\mu_b \leqslant \frac{(\eta_L+\mu_L)\mu_m}{\eta_L}$, then the *DFE* is $(N_h, 0, 0, 0, 0, 0, 0)$, i.e. an equilibrium without mosquitoes, which is not biologically realistic. If $\mu_b > \frac{(\eta_L+\mu_L)\mu_m}{\eta_L}$, the *DFE* is $(N_h, 0, 0, L_{m0}, S_{m0}, 0, 0)$, which is biologically realistic.

Is it possible to find an equilibrium such that $I_h^* > 0$ and $I_m^* > 0$? We consider $L_m = L_{m0} = \frac{\mu_m}{\eta_L} S_{m0}$. Then, using $(2)_2$, we have

$$S_m^* = \frac{\mu_m S_{m0}}{\mu_m + C\beta_{hm} \frac{I_h^*}{N_h}}$$

and we deduce that

$$\begin{cases} S_{m0} = S_m^* + \frac{\mu_m}{\eta_m} I_m^* + I_m^* \\ I_m^* = \frac{\eta_m}{\mu_m+\eta_m}(S_{m0} - S_m^*) \\ \quad = \frac{\eta_m}{\mu_m+\eta_m}\left(S_{m0} - \frac{\mu_m S_{m0}}{\mu_m + C\beta_{hm}\frac{I_h^*}{N_h}}\right). \\ \quad = \frac{\eta_m}{\mu_m+\eta_m} \frac{C\beta_{hm}\frac{I_h^*}{N_h}}{\mu_m + C\beta_{hm}\frac{I_h^*}{N_h}} S_{m0} \end{cases}$$

Finally, using $(12)_4$ with $I_h = I_h^*$ and $I_m = I_m^*$, we obtain

$$\frac{I_h^*}{N_h} = \frac{\nu_h \mu_h B \beta_{mh}}{(\mu_h+\nu_h)(\mu_h+\eta_h)(\mu_h N_h + B\beta_{mh} I_m^*)} \frac{\eta_m}{\mu_m+\eta_m} \times \frac{C\beta_{hm}\frac{I_h^*}{N_h}}{\mu_m + C\beta_{hm}\frac{I_h^*}{N_h}} S_{m0}.$$

$I_h > 0$, we deduce

$$\begin{aligned} &(\mu_h N_h + B\beta_{mh} I_m^*)\left(\mu_m + C\beta_{hm}\frac{I_h^*}{N_h}\right) \\ &= \frac{\nu_h \mu_h \eta_m B \beta_{mh} C \beta_{hm}}{(\mu_h+\nu_h)(\mu_h+\eta_h)(\mu_m+\eta_m)} S_{m0}. \\ &B\beta_{mh} I_m^* \left(\mu_m + C\beta_{hm}\frac{I_h^*}{N_h}\right) + \mu_h N_h C\beta_{hm}\frac{I_h^*}{N_h} \\ &= \frac{\nu_h \mu_h \eta_m B \beta_{mh} C \beta_{hm}}{(\mu_h+\nu_h)(\mu_h+\eta_h)(\mu_m+\eta_m)} S_{m0} - \mu_m \mu_h N_h \\ &= \mu_m \mu_h N_h (R_0^2 - 1). \end{aligned}$$

The left-hand side is positive if and only $R_0^2 > 1$. Then, we deduce that

$$\left(\frac{\eta_m}{\mu_m+\eta_m} B\beta_{mh} C\beta_{hm} S_{m0} + \mu_h N_h C\beta_{hm}\right)\frac{I_h^*}{N_h} = \mu_m \mu_h N_h (R_0^2 - 1)$$

$$I_h^* = \frac{\mu_m \mu_h N_h (R_0^2 - 1)}{C\beta_{hm}\left(\frac{\eta_m}{\mu_m+\eta_m} B\beta_{mh} S_{m0}/N_h + \mu_h\right)} > 0$$

if and only if $R_0^2 > 1$. Then, using some of the relations above, we deduce $E_h^* > 0$, $S_h^* > 0$, $R_h^* > 0$, $I_m^* > 0$, $S_m^* > 0$, $E_m^* > 0$. Finally, there exists a unique endemic equilibrium $(S_h^*, E_h^*, I_h^*, R_h^*, L_{m0}, S_m^*, E_m^*, I_m^*)$ if and only if $R_0^2 > 1$.

Suppose now $\mu_b \leqslant \frac{(\eta_L+\mu_L)\mu_m}{\eta_L}$. For $(N_h, 0, 0, 0, 0, 0, 0)$, the Jacobian matrix is given by

$$J_{DFE} = \begin{pmatrix} -\mu_h & 0 & 0 & 0 & 0 & 0 & -B\beta_{mh} \\ 0 & -(\nu_h+\mu_h) & 0 & 0 & 0 & 0 & B\beta_{mh} \\ 0 & \nu_h & -(\mu_h+\eta_h) & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -(\mu_L+\eta_L) & \mu_b & \mu_b & \mu_b \\ 0 & 0 & & \eta_L & -\mu_m & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -(\mu_m+\eta_m) & 0 \\ 0 & 0 & 0 & 0 & 0 & \eta_m & -\mu_m \end{pmatrix}$$

from which we deduce the characteristic polynomial

$$p(\lambda) = -(\lambda+\mu_h)(\lambda+\mu_h+v_h)(\lambda+\mu_h+\eta_h)(\lambda+\mu_m+\eta_m) \times (\lambda+\mu_m)q(\lambda),$$

with $q(\lambda) = (\lambda+\mu_L+\eta_L)(\lambda+\mu_m)+\eta_L\mu_b$. The eigenvalues of matrix J_{DFE} are $-\mu_h$, $-(\mu_h+v_h)$, $-(\mu_h+\eta_h)$, $-\mu_m$, $-(\mu_m+\eta_m)$ and the roots of q. The coefficients of q are positive, which implies that the roots of q have a negative real part. Thus, if $\mu_b \leqslant \frac{(\eta_L+\mu_L)\mu_m}{\eta_L}$, the state $(N_h,0,0,0,0,0,0)$ is locally asymptotically stable.

Appendix B

We compute the Jacobian matrix associated with $\mathcal{F}$ and $\mathcal{V}$. A direct computation gives

$$J_{\mathcal{V}}(x) = \begin{pmatrix} v_h+\mu_h & 0 & 0 & 0 \\ -v_h & \eta_h+\mu_h & 0 & 0 \\ 0 & 0 & \mu_m+\eta_m & 0 \\ 0 & 0 & -\eta_m & \mu_m \end{pmatrix},$$

$$J_{\mathcal{F}}(x) = \begin{pmatrix} 0 & 0 & 0 & B\beta_{mh}\frac{S_h}{N_h} \\ 0 & 0 & 0 & 0 \\ 0 & C\beta_{hm}\frac{S_m}{N_h} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Then computing $J_{\mathcal{V}}^{-1}$, we deduce at x_{DFE}:

$$J_{\mathcal{F}}(x_{DFE})J_{\mathcal{V}}^{-1}(x_{DFE}) = \begin{pmatrix} 0 & 0 & B\beta_{mh}\frac{\eta_m}{\mu_m(\mu_m+\eta_m)} & B\beta_{mh}\frac{v_m}{\mu_m} \\ 0 & 0 & 0 & 0 \\ C\beta_{hm}\frac{v_h}{(\mu_h+\eta_h)(\mu_h+v_h)}\frac{S_{m0}}{N_h} & C\beta_{mh}\frac{v_h}{\mu_h+v_h} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

We deduce the characteristic polynomial

$$p(\lambda) = \lambda^2\left(\lambda^2 - B\beta_{mh}\frac{\eta_m}{\mu_m(\mu_m+\eta_m)}C\beta_{hm}\frac{v_h}{(\mu_h+v_h)(\mu_h+\eta_h)}\frac{S_{m0}}{N_h}\right).$$

Finally, we have $\rho(J_{\mathcal{F}}J_{\mathcal{V}}^{-1}) = \sqrt{\frac{\eta_m B\beta_{mh}C\beta_{hm}v_h}{\mu_m(\mu_m+\eta_m)(\mu_h+v_h)(\mu_h+\eta_h)}\frac{S_{m0}}{N_h}}$.

Appendix C

Here, we study the eigenvalues of the Jacobian matrix associated with the right-hand side of the numerical algorithm (10)–(11) at the *DFE*, without the equation for R_h^n to lower the size of the matrix. We obtain

$$J_{DFE} = \begin{pmatrix} \frac{1}{1+\Delta t\mu_h} & 0 & 0 & 0 & 0 & 0 & \frac{-\Delta tB\beta_{mh}}{1+\Delta t\mu_h} \\ 0 & \frac{1}{1+\Delta t(v_h+\mu_h)} & 0 & 0 & 0 & 0 & \frac{\Delta tB\beta_{mh}}{1+\Delta t(v_h+\mu_h)} \\ 0 & \frac{\Delta tv_h}{1+\Delta t(\mu_h+\eta_h)} & \frac{1}{1+\Delta t(\mu_h+\eta_h)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{1+\Delta t\mu_b\eta_L/\mu_m} & \frac{\Delta t(\mu_L+\eta_L)}{\mu_L(1+\Delta t\mu_b\eta_L/\mu_m)} & \frac{\Delta t(\mu_L+\eta_L)}{\mu_L(1+\Delta t\mu_b\eta_L/\mu_m)} & \frac{\Delta t(\mu_L+\eta_L)}{\mu_L(1+\Delta t\mu_b\eta_L/\mu_m)} \\ 0 & 0 & -\frac{\Delta tC\beta_{hm}}{1+\Delta t(\mu_m+\eta_m)}\frac{S_{m0}}{N_h} & \frac{\Delta t\mu_L}{1+\Delta t\mu_L} & \frac{1}{1+\Delta t\mu_L} & 0 & 0 \\ 0 & 0 & \frac{\Delta tC\beta_{hm}}{1+\Delta t(\mu_m+\eta_m)}\frac{S_{m0}}{N_h} & 0 & 0 & \frac{1}{1+\Delta t(\mu_m+\eta_m)} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\Delta t\eta_m}{1+\Delta t\mu_m} & \frac{1}{1+\Delta t\mu_m}. \end{pmatrix}$$

After long and tedious computations, we obtain the characteristic polynomial associated with J_{DFE}

$$p(\lambda) = \left(\frac{1}{1+\Delta t\mu_h}-\lambda\right) * q_1(\lambda) * q_2(\lambda)$$

where q_1 and q_2 are defined as follows

$$q_1(\lambda) = \left(\frac{1}{1+\Delta t\mu_b\eta_L/\mu_m}-\lambda\right)\left(\frac{1}{1+\Delta t\mu_L}-\lambda\right) + \frac{\Delta t(\mu_L+\eta_L)\mu_m}{\mu_L(1+\Delta t\mu_b\eta_L/\mu_m)}\frac{\Delta t\mu_L}{1+\Delta t\mu_L}$$

$$q_2(\lambda) = \left(\frac{1}{1+\Delta t\mu_m}-\lambda\right)\left(\frac{1}{1+\Delta t(\mu_m+\eta_m)}-\lambda\right) \times \left(\frac{1}{1+\Delta t(\eta_h+\mu_h)}-\lambda\right)\left(\frac{1}{1+\Delta t(v_h+\mu_h)}-\lambda\right) + \frac{\Delta t^4CB\beta_{hm}\beta_{mh}\eta_m v_h}{(1+\Delta t\mu_m)(1+\Delta t(\mu_m+\eta_m))(1+\Delta t(\eta_h+\mu_h))(1+\Delta t(v_h+\mu_h))} \times \frac{S_{m0}}{N_h}.$$

We need to show that each root λ_i verifies $|\lambda_i| < 1$. It is clear that $\lambda_1 = \frac{1}{1+\Delta t\mu_h} < 1$. We need also to show that q_1 and q_2 are Schur polynomials, i.e. polynomials such that all roots λ_i verify $|\lambda_i| < 1$.

Remark 10. If we consider the whole difference system (10,11), we obtain the following characteristic polynomial

$$p(\lambda) = \left(\frac{1}{1+\Delta t\mu_h}-\lambda\right)^2 * q_1(\lambda) * q_2(\lambda),$$

instead of the previous one.

In the case of q_1, we will use the following nice result.

Lemma 11. *Let $q(x) = x^2 + a_1x + a_2$ with $a_i \in \mathbb{R}$. Then each root λ_i verifies $|\lambda_i| < 1$ if and only if*

$$1 + a_1 + a_2 > 0, \qquad 1 - a_1 + a_2 > 0, \qquad 1 - a_2 > 0.$$

We have

$$q_1(\lambda) = \lambda^2 - \lambda\left(\frac{1}{1+\Delta t\mu_b\eta_L/\mu_m} + \frac{1}{1+\Delta t\mu_L}\right) + \frac{1-\Delta t^2(\mu_L+\eta_L)\mu_m}{(1+\Delta t\mu_b\eta_L/\mu_m)(1+\Delta t\mu_L)}.$$

Setting $a_1 = -(\frac{1}{1+\Delta t\mu_b\eta_L/\mu_m} + \frac{1}{1+\Delta t\mu_L})$ and $a_2 = \frac{1+\Delta t^2(\mu_L+\eta_L)\mu_m}{(1+\Delta t\mu_b\eta_L/\mu_m)(1+\Delta t\mu_L)}$, we compute

$$1-a_2 = \left(1+\Delta t\frac{\mu_b\eta_L}{\mu_m}\right)(1+\Delta t\mu_L) - (1+\Delta t^2(\mu_L+\eta_L)\mu_m) = \frac{\mu_b\eta_L}{\mu_m} + \mu_L + \Delta t\left(\frac{\mu_b\eta_L\mu_L}{\mu_m} - (\mu_L+\eta_L)\mu_m\right) > 0.$$

We know that μ_b is chosen such that $\frac{\mu_b\eta_L}{\mu_m} - (\mu_L+\eta_L) \geqslant 0$. Thus the previous inequality becomes $\mu_L > \mu_m$, which means that the mortality rate in the aquatic stage is greater than the mortality rate for adult mosquitoes. Thus, up to $\mu_L > \mu_m$, we have $1-a_2 > 0$ for all $\Delta t > 0$. Then

$$1+a_1+a_2 = \left(1+\Delta t\frac{\mu_b\eta_L}{\mu_m}\right)(1+\Delta t\mu_L) - \left(\left(1+\Delta t\frac{\mu_b\eta_L}{\mu_m}\right) + (1+\Delta t\mu_L)\right) + (1+\Delta t^2(\mu_L+\eta_L)\mu_m), = \frac{\mu_b\eta_L\mu_L}{\mu_m} + (\mu_L+\eta_L)\mu_m > 0.$$

Because $a_1 < 0$, we deduce that $1-a_1+a_2 > 1+a_1+a_2 > 0$ for all $\Delta t > 0$.

Finally, we conclude that if $\mu_L \geqslant \mu_m$, the roots of q_1 have a modulus lower than 1, for all $\Delta t > 0$.

We turn now to the fourth-order polynomial q_2. Using the change of variable $\lambda = \frac{1+z}{1-z}$ and a Routh–Hurwitz criterion, we show that $(1-z)^4 q_2(\frac{1+z}{1-z})$ has all its roots in the negative Real plane, which implies that the roots of $q_2(\lambda)$ have a modulus lower than 1. Our polynomial q_2 can be rewritten in the following way

$$q_2(\lambda) = \left(\frac{1}{a+1}-\lambda\right)\left(\frac{1}{b+1}-\lambda\right)\left(\frac{1}{c+1}-\lambda\right)\left(\frac{1}{d+1}-\lambda\right) + \frac{e}{(a+1)(b+1)(c+1)(d+1)} = \frac{1}{(a+1)(b+1)(c+1)(d+1)} \times ((1-(a+1)\lambda)(1-(b+1)\lambda) \times (1-(c+1)\lambda)(1-(d+1)\lambda)+e),$$

with $a = \Delta t\mu_m$, $b = \Delta t(\mu_m+\eta_m)$, $c = \Delta t(\mu_h+\nu_h)$, $d = \Delta t(\mu_h+\eta_h)$ and $e = \Delta t^4 CB\beta_{hm}\beta_{mh}\eta_m\nu_h\frac{S_{m0}}{N_h} = abcdR_0^2$. Thus

$$(1-z)^4 q_2\left(\frac{1+z}{1-z}\right) = \frac{((a+2)z+a)((b+2)z+b)((c+2)z+c)((d+2)z+d)+e(1-z)^4}{(a+1)(b+1)(c+1)(d+1)} = A_0z^4 + A_1z^3 + A_2z^2 + A_3z + A_4,$$

with

$$A_0 = e + (a+2)(b+2)(c+2)(d+2) > 0,$$
$$A_1 = -4e + (a+2)(b+2)(c(d+2)+d(c+2)) + (c+2)*(d+2)(a(b+2)+b(a+2)) > 4(abcd-e) = 4abcd(1-R_0^2),$$
$$A_2 = 6e + (a+2)(b+2)cd + (c+2)(d+2)ab + 2(a+ab+b)(c+cd+d) > 0,$$
$$A_3 = -4e + (a+2)bcd + (b+2)acd + (c+2)abd + (d+2)abc > 4(abcd-e) > 4abcd(1-R_0^2),$$
$$A_4 = e + abcd > 0.$$

Thus, it is clear that all coefficients are always positive since $R_0 < 1$. Finally, we just have to verify that

$$A_1A_2A_3 - A_0A_3^2 - A_4A_2^2 > 0, \tag{13}$$

which is a rather complicated inequality to study, at least theoretically. In fact, with the parameters of Table 1 we numerically checked that the previous inequality was always true in our simulations for all $\Delta t > 0$.

References

[1] R.M. Anderson, R.M. May, Infectious Diseases of Humans: Dynamics and Control, Oxford university Press, Oxford, 1991.

[2] R. Anguelov, J.M-S. Lubuma, Contributions to the mathematics of the nonstandard finite difference method and applications, Numer. Methods Partial Diff. Eq. 17 (2001) 518.

[3] N. Bacaer, Approximation of the basic reproduction number R_0 for vector-borne diseases with a periodic vector population, Bull. Math. Bio. 69 (2007) 1067.

[4] M.Q. Benedict, R.S. Levine, W.A. Hawley, L.P. Lounibo, Spread of the tiger: global risk of invasion by the mosquito *Aedes albopictus*, Vector-Borne Zoonotic Dis 7 (2007) 76.

[5] L.E. Campos, A. San Juan, L.C. Cenabre, E.F. Almagro, Isolation of chikungunya virus in the Philippines, Acta Med. Philippina 5 (1969) 152.

[6] D.E. Carey, Chikungunya and dengue: a case of mistaken identity? J. Hist. Med. Allied Sci. 26 (1971) 243.

[7] D.E. Carey, R.M. Myers, C.M. Deranitz, M. Jadhav, R. Reuben, The 1964 Chikungunya epidemic at Vellore, South India, including observations on concurrent dengue, Trans. R. Soc. Trop. Med. Hyg. 63 (4) (1969) 434.

[8] C. Chastel, Infections humaines au Cambodge par le virus Chikungunya ou un agent étroitement apparenté. II Anatomie pathologique expérimentale, Bull. Soc. Pathol. Exot. 56 (1963) 915.

[9] S.R. Christopher, *Aedes aegypti*, the Yellow fever mosquitoIts Life History, Bionomics and Structure, Cambridge University Press, Cambridge, 1960.

[10] M. Derouich, A. Boutayeb, E.H. Twizell, A model of Dengue fever, BioMed. Eng. Online 2 (2003).

[11] M. Diallo, J. Thonnon, M. Traore-Lamizana, D. Fontenille, Vectors of Chikungunya virus in Senegal: current data and transmission cycles, Am. J. Trop. Med. Hyg. 60 (1999) 281.

[12] O. Diekman, J.A.P. Heesterbeck, Mathematical epidemiology of infectious diseasesWiley Series in Mathematical and Computational Biology, Wiley, New York, 2000.

[13] L. Esteva, C. Vargas, Analysis of a Dengue disease transmission model, Math. Biosci. 150 (1998) 131.

[14] L. Esteva, C. Vargas, A model for Dengue disease with variable human population, J. Math. Biol. vol. 38 (1999) 220.

[15] A. Failloux, Personal communication, 2007.

[16] W.H. Hawley, The biology of *Aedes albopictus*, J. Am. Mosquito Control Assoc. 4 (1988) 1.
[17] J.M. Hefferman, R.J. Smith, L.M. Wahl, Perspectives on the basic reproductive number, J. R. Soc. Interf. 2 (2005) 281.
[18] H.W. Hethcote, The mathematics of infectious diseases, SIAM Rev. 42 (2000) 599.
[19] J.C. Kamgang, G. Sallet, Global asymptotic stability for the disease free equilibrium for epidemiological models, CRAS Ser. I 341 (2005) 433.
[20] Kass' Moustik, <www.chikungunya.gouv.fr>.
[21] F.F. Macasaet, Further observation on chikungunya fever, J. Philippines Med. Assoc. 46 (1970) 235.
[22] J.S. Mackenzie, K.B. Chua, P.W. Daniels, B.T. Eaton, H.E. Field, R.A. Hall, et al., Emerging viral diseases of Southeast Asia and the Western Pacific, Emerg. Infect. Dis. 7 (2001) 497.
[23] J.M. Medlock, D. Avenell, I. Barrass, S. Leach, Analysis of the potential for survival and seasonal activity of *Aedes albopictus* (Diptera: *Culicidae*) in the United Kingdom, J. Vector Ecol. 31 (2006) 292.
[24] R.E. Mickens, Nonstandard Finite Difference Models of Differential Equations, World Scientific, Singapore, 1994.
[25] R.E. Mickens, Applications of Nonstandard Finite Difference Schemes, World Scientific, Singapore, 2000.
[26] R.E. Mickens, Nonstandard finite difference schemes for differential equations, J. Diff. Eq. Appl. 8 (2002) 823.
[27] R.E. Mickens (Ed.), Advances in the Applications of Nonstandard Finite Difference Schemes, World Scientific, Singapore, 2005.
[28] R.M. Myers, D.E. Carey, R. Reuben, E. Jesudass, C. Deranitz, J. Malati, The 1964 epidemic of dengue-like fever in South India. Isolation of Chikungunya virus from human sera and from mosquitoes, Indian J. Med. Res. 53 (1965) 694.
[29] G. Pialoux, B.-A. Gaüzere, S. Jauréguiberry, M. Strobel, Chikungunya, an epidemic arbovirosis, Lancet Infect. Dis. 7 (2007) 319.
[30] R.W. Ross, The Newala epidemic. III. The virus: isolation, pathogenic properties and relationship to the epidemic, J. Hyg. 54 (1956) 177.
[31] P. Van den Driessche, J. Watmough, Reproduction numbers and subthreshold endemic equilibria for compartmental models of disease transmission, Math. Biosci. 180 (2002) 29.